

Source-route qualification matrix (bounded S1 review)

	Route 1 COMSOL 6.4	Route 2 PCMO reaction-drift	Route 3 SYN_EDT_2D_V1
Research-use / executable rights	NOT ESTABLISHED	UNKNOWN / NOT PASS	N/A — synthetic
2D device-field topology	DOCUMENTED	FAIL — point device	ENGINEERING-SPECIFIED
Conservative defect field	IDENTITY-SCOPED	FAIL	ENGINEERING-SPECIFIED
Complete IC / BC / interfaces	PARTIAL	FAIL	ENGINEERING-SPECIFIED
Absolute waveform	DOCUMENTED	FAIL after compliance	ENGINEERING-SPECIFIED
Machine-readable reference fields	FAIL	FAIL	LOCAL OUTPUTS ONLY*
S1 route disposition	FAIL	FAIL	ACTIVATED AT S1*

* Route 3 was activated as a non-source-aligned benchmark; S2 later terminated numerically (Fig. 3).